

template

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https://github.com/stellassymonds/ENVS-193DS_spring-2026_final

```
# read in packages
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(DHARMA)

# read in data
nests <- read.csv(here("data", "nest_data_final.csv"))
```

Problem 1

- a. Part 1: Part 2:
- b. Part 1: We found evidence to suggest that the distance to water edge does predict probability of American Avocet microhabitat use.

Part 2: We found evidence to suggest that there is a relationship between American Avocet, Forster's Tern, and Black-necked Stilt bird species and managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat habitats.

- c. Associated behavior type of either feeding, breeding, or roosting could be a categorical variable that influences the probability of American Avocet microhabitat use. For example, waterbirds that are breeding or roosting may prefer islands within ponds.

Problem 2

- a.
- b. 1s mean x and 0s mean y.
- c. Tree height is a X variable. Acacia ant species is a Y variable.
- d.
- e.
- f.
- g.
- h.
- i. Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>
- j.
- k.

Problem 3

- a. How are the visualizations different from each other in the way you have represented your data?

What similarities do you see between all your visualizations?

What patterns (e.g. differences in means/counts/proportions/medians, trends through time, relationships between variables) do you see in each visualization? Are these different between visualizations? If so, why? If not, why not? A slight difference in medians between bike rides when I have plans after class and no plans after class is visible in my exploratory visualization,

My feedback in workshop consisted of possibly adding icons to the end of the lines to represent temperature or another variable. I ultimately decided to not implement this feedback, since neither temperature nor other variables I tracked had any visible affect on my data, and I thought it may make it harder to see the difference between the lengths of each stitched line.

- b. This analysis is appropriate to find a correlation between status of my plans after class and my bike ride durations. one variable is binary
- c.